

Pragati Engineering College



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Carbon Footprint Calculator

A console-based C++ application to estimate annual CO2 emissions across Transport, Diet, and Home Energy — aligned with UN SDG Goal 13: Climate Action.

Project Description

This project is a console-based **Carbon Footprint Calculator** built in C++ that helps individuals estimate their annual CO2 emissions across three key lifestyle categories: **Transport**, **Diet**, and **Home Energy**. The program applies real-world Indian emission factors to produce accurate kg CO2/year values.

The application compares the user's total footprint against Indian (1,900 kg/year) and global (4,000 kg/year) averages, provides a status verdict, generates personalised reduction suggestions, and optionally saves a formatted report to **carbon_report.txt** using file I/O.

Key Features

- **3 OOP Classes:** TransportData, DietData, HomeEnergyData — each with getInput() and calculateEmissions() methods.
- **Emission Factor Constants:** #define macros for Bus (0.089), Bike (0.103), Train (0.041), LPG (2.983), Electricity (0.82 kg CO2/kWh).
- **Real-world Benchmarks:** Compares result against Indian average (1,900 kg/yr) and global average (4,000 kg/yr).
- **Status Verdict:** Excellent / Moderate / High footprint classification based on comparison.
- **Personalised Suggestions:** Rule-based tips on transport, diet, and energy with calculated potential savings.
- **File Report Export:** Saves full breakdown to carbon_report.txt using ofstream.
- **Input Validation:** Structured menu-driven input with numeric selection for diet type.
- **Formatted Output:** Uses iomanip (setw, setprecision, fixed) for clean console display.

Technologies Used

Technology / Feature	Purpose in This Project
C++ (Standard)	Core language — OOP classes, functions, macros
iostream	Console input/output (cin, cout)
iomanip	Formatted output: fixed, setprecision, setw

fstream (ofstream)	File I/O — saves report to carbon_report.txt
#define Macros	Emission factor constants (BUS_PER_KM, LPG_PER_KG, etc.)
OOP — 3 Classes	TransportData, DietData, HomeEnergyData
Functions	printLine(), printHeading(), showResults(), saveReport()
if-else / switch	Diet selection, status verdict, suggestion logic

How to Run the Project

All Operating Systems — Windows · macOS · Linux

Step 1: Open Terminal / Command Prompt

Windows: Use **Command Prompt** or **PowerShell**
macOS / Linux: Open the default **Terminal** application.

Step 2: Navigate to the Project Folder

Use the `cd` command to enter the project directory.
Example: `cd Desktop/Carbon_Footprint_Project`

Step 3: Compile the Program

`g++ Carbon_Footprint_Calculator.cpp -o Carbon_Footprint_Calculator`
This compiles using GCC. Works on Windows (MinGW), macOS, and Linux.

Step 4: Run the Program

macOS / Linux: `./Carbon_Footprint_Calculator`
Windows: `Carbon_Footprint_Calculator.exe`

Files in the Project

File	Description
Carbon_Footprint_Calculator.cpp	Main source file — all classes, functions, and <code>main()</code>
carbon_report.txt	Auto-generated report file saved when user selects Yes
README.md	Project documentation and usage guide

Important Notes

- **Emission Factors:** Indian-context values — Electricity: 0.82 kg CO₂/kWh (India grid); LPG: 2.983 kg CO₂/kg; Bus: 0.089, Bike: 0.103, Train: 0.041 kg CO₂/km.
- **Diet Units:** Diet emissions are in kg CO₂ *per day* multiplied by 365 for annual total.
- **Transport:** Weekly km inputs are multiplied by 52 (weeks/year) for annual values.
- **File Output:** carbon_report.txt is created or overwritten in the working directory each run.
- **Compiler:** Any standard C++ compiler (g++, clang++, MSVC). No external libraries needed.

Section Explanations

3 Modules — Transport · Diet · Home Energy

Section 1: Transport Emissions — TransportData Class

Calculates annual CO2 from weekly travel by bus, bike, and train.

Parameter / Factor	Value / Detail
Input	Bus km/week, Bike km/week, Train km/week
Formula	Annual CO2 = (km/week × 52) × emission_factor for each mode
Bus Factor	0.089 kg CO2/km (BUS_PER_KM)
Bike Factor	0.103 kg CO2/km (BIKE_PER_KM)
Train Factor	0.041 kg CO2/km (TRAIN_PER_KM)
Example	20 km/week bus → 20 × 52 × 0.089 = 92.56 kg CO2/year
Class Method	getInput() collects data; calculateEmissions() returns total

Section 2: Diet Emissions — DietData Class

Estimates annual CO2 from dietary choices using daily emission rates.

Parameter / Factor	Value / Detail
Input	Diet type: 1 = Non-Vegetarian, 2 = Average Mixed, 3 = Vegetarian
Non-Vegetarian	3.3 kg CO2/day → 3.3 × 365 = 1,204.5 kg CO2/year
Average Mixed	2.5 kg CO2/day → 2.5 × 365 = 912.5 kg CO2/year
Vegetarian	1.7 kg CO2/day → 1.7 × 365 = 620.5 kg CO2/year
Default	If no valid choice, defaults to Average Mixed Diet
Class Method	getInput() shows menu; calculateEmissions() returns annual value

Section 3: Home Energy Emissions — HomeEnergyData Class

Calculates annual CO2 from monthly electricity and LPG gas consumption.

Parameter / Factor	Value / Detail
Input	Electricity kWh/month, LPG kg/month
Electricity Factor	0.82 kg CO2/kWh (ELECTRICITY_PER_KWH — India grid factor)
LPG Factor	2.983 kg CO2/kg (LPG_PER_KG)
Formula	Annual = (elec_kWh × 12 × 0.82) + (lpg_kg × 12 × 2.983)
Example	150 kWh/month → 150 × 12 × 0.82 = 1,476 kg CO2/year
Class Method	getInput() collects usage; calculateEmissions() returns total

Results, Suggestions & File Output

showResults() · saveReport() — Logic Breakdown

showResults() — Status & Suggestions Logic

Condition	Comparison	Action / Output
total < 1,900 kg	< Indian Average	Status: Excellent! Below Indian average.
1,900–4,000 kg	Between averages	Status: Moderate footprint.
> 4,000 kg	> Global Average	Status: High carbon footprint.
transport > 500 kg	Transport trigger	Tip: Use public transport more often. Save 20%.
diet > veg × 365	Diet trigger	Tip: Reduce meat consumption. Shows exact savings.
energy > 800 kg	Energy trigger	Tip: Use energy-efficient appliances. Save 20%.

saveReport() — File Output Format

```
=====
CARBON FOOTPRINT REPORT
=====
Name       : Mohith
Transport  : 449.28 kg C02/year
Diet       : 912.50 kg C02/year
Home Energy : 2,686.80 kg C02/year
-----
Total      : 4048.58 kg C02/year
=====
```

Saved to: **carbon_report.txt** using ofstream in append/overwrite mode.

Source Code — Part 1

Headers, Macros, Helper Functions & Classes

Headers, Macros & Helper Functions

```
#include <iostream>
#include <string>
#include <iomanip>
#include <fstream>
using namespace std;

// Emission Factor Constants (Indian context)
#define BUS_PER_KM      0.089    // kg CO2 per km
#define BIKE_PER_KM     0.103
#define TRAIN_PER_KM    0.041
#define DIET_NONVEG     3.3      // kg CO2 per day
#define DIET_AVERAGE    2.5
#define DIET_VEGETARIAN 1.7
#define ELECTRICITY_PER_KWH 0.82 // kg CO2 per kWh (India grid)
#define LPG_PER_KG       2.983 // kg CO2 per kg

void printLine(char symbol = '-') {
    for (int i = 0; i < 60; i++)
        cout << symbol;
    cout << endl;
}

void printHeading(string title) {
    cout << endl;
    printLine('=');
    cout << ' ' << title << endl;
    printLine('=');
}
```

TransportData Class

```
class TransportData {
private:
    double busKmPerWeek;
    double bikeKmPerWeek;
    double trainKmPerWeek;
public:
    TransportData() {
        busKmPerWeek = 0; bikeKmPerWeek = 0; trainKmPerWeek = 0;
    }
    void getInput() {
        printHeading("SECTION 1 : TRANSPORT");
        cout << "\nBus travel per week (km)   : "; cin >> busKmPerWeek;
        cout << "Bike travel per week (km)  : "; cin >> bikeKmPerWeek;
        cout << "Train travel per week (km) : "; cin >> trainKmPerWeek;
    }
    double calculateEmissions() {
        double bus    = busKmPerWeek    * 52 * BUS_PER_KM;
        double bike    = bikeKmPerWeek   * 52 * BIKE_PER_KM;
        double train   = trainKmPerWeek  * 52 * TRAIN_PER_KM;
        return bus + bike + train;
    }
};
```

DietData & HomeEnergyData Classes

```
class DietData {
private:
    int dietType;
public:
    DietData() { dietType = 2; }
    void getInput() {
        printHeading("SECTION 2 : DIET");
        cout << "\n1. Non-Vegetarian" << endl;
        cout << "2. Average Mixed Diet" << endl;
        cout << "3. Vegetarian" << endl;
        cout << "\nEnter your choice : "; cin >> dietType;
    }
    double calculateEmissions() {
        double dailyEmission = DIET_AVERAGE;
        if (dietType == 1)        dailyEmission = DIET_NONVEG;
        else if (dietType == 3) dailyEmission = DIET_VEGETARIAN;
        return dailyEmission * 365;
    }
};

class HomeEnergyData {
private:
    double electricityUnitsPerMonth;
    double lpgKgPerMonth;
public:
    HomeEnergyData() { electricityUnitsPerMonth = 0; lpgKgPerMonth = 0; }
    void getInput() {
        printHeading("SECTION 3 : HOME ENERGY");
        cout << "\nElectricity usage per month (kWh) : ";
        cin >> electricityUnitsPerMonth;
        cout << "LPG usage per month (kg)          : ";
        cin >> lpgKgPerMonth;
    }
    double calculateEmissions() {
        double electricity = electricityUnitsPerMonth * 12 * ELECTRICITY_PER_KWH;
        double lpg          = lpgKgPerMonth * 12 * LPG_PER_KG;
        return electricity + lpg;
    }
};
```

Source Code — Part 2

showResults(), saveReport() & main()

showResults() Function

```
void showResults(double transport, double diet, double energy, string name) {
    double total = transport + diet + energy;
    double avgIndian = 1900;
    double avgGlobal = 4000;

    printHeading("CARBON FOOTPRINT RESULTS");
    cout << fixed << setprecision(2);
    cout << "\nName : " << name << endl;
    printLine();
    cout << "Transport Emissions : " << setw(10) << transport << " kg CO2/year" << endl;
    cout << "Diet Emissions      : " << setw(10) << diet      << " kg CO2/year" << endl;
    cout << "Home Energy           : " << setw(10) << energy    << " kg CO2/year" << endl;
    printLine();
    cout << "Total Footprint       : " << setw(10) << total      << " kg CO2/year" << endl;
    printLine();
    cout << "\nIndian Average : " << avgIndian << " kg CO2/year" << endl;
    cout << "Global Average : " << avgGlobal << " kg CO2/year" << endl;

    if      (total < avgIndian) cout << "\nStatus : Excellent! Below Indian average." << endl;
    else if (total < avgGlobal) cout << "\nStatus : Moderate footprint." << endl;
    else      cout << "\nStatus : High carbon footprint." << endl;

    printHeading("SUGGESTIONS");
    if (transport > 500) {
        cout << "\nUse public transport more often." << endl;
        cout << "Possible reduction : " << transport * 0.20 << " kg CO2/year" << endl;
    }
    if (diet > DIET_VEGETARIAN * 365) {
        cout << "\nReduce meat consumption." << endl;
        cout << "Potential savings : " << (diet - DIET_VEGETARIAN*365) << " kg CO2/year" << endl;
    }
    if (energy > 800) {
        cout << "\nUse energy-efficient appliances." << endl;
        cout << "Possible reduction : " << energy * 0.20 << " kg CO2/year" << endl;
    }
    if (total < avgIndian)
        cout << "\nGreat job maintaining a low footprint!" << endl;
}
```

saveReport() & main()

```

void saveReport(string name, double transport, double diet, double energy) {
    double total = transport + diet + energy;
    ofstream file("carbon_report.txt");
    if (!file) { cout << "\nUnable to save report." << endl; return; }
    file << fixed << setprecision(2);
    file << "=====\n";
    file << " CARBON FOOTPRINT REPORT\n";
    file << "=====\n";
    file << "Name      : " << name      << "\n";
    file << "Transport  : " << transport << " kg CO2/year\n";
    file << "Diet       : " << diet       << " kg CO2/year\n";
    file << "Home Energy : " << energy    << " kg CO2/year\n";
    file << "-----\n";
    file << "Total      : " << total      << " kg CO2/year\n";
    file << "=====\n";
    file.close();
    cout << "\nReport saved successfully." << endl;
}

int main() {
    printLine('*');
    cout << " CARBON FOOTPRINT CALCULATOR" << endl;
    printLine('*');
    string userName;
    cout << "\nEnter your name : "; cin >> userName;

    TransportData transport;
    DietData      diet;
    HomeEnergyData energy;

    transport.getInput();
    diet.getInput();
    energy.getInput();

    double tE = transport.calculateEmissions();
    double dE = diet.calculateEmissions();
    double eE = energy.calculateEmissions();

    showResults(tE, dE, eE, userName);

    int choice;
    cout << "\nSave report to file? (1 = Yes, 0 = No) : ";
    cin >> choice;
    if (choice == 1) saveReport(userName, tE, dE, eE);

    printLine('*');
    cout << " Thank you for using the calculator!" << endl;
    printLine('*');
    return 0;
}

```

Sample Output

Full Console Session — Player: Mohith

```
*****
CARBON FOOTPRINT CALCULATOR
*****

Enter your name : Mohith

=====

SECTION 1 : TRANSPORT
=====

Bus travel per week (km)   : 30
Bike travel per week (km)  : 10
Train travel per week (km) : 20

=====

SECTION 2 : DIET
=====

1. Non-Vegetarian
2. Average Mixed Diet
3. Vegetarian

Enter your choice : 2
```

```
=====
SECTION 3 : HOME ENERGY
=====

Electricity usage per month (kWh) : 200
LPG usage per month (kg)          : 12

=====

CARBON FOOTPRINT RESULTS
=====

Name : Mohith
-----
Transport Emissions :    299.52 kg CO2/year
Diet Emissions      :    912.50 kg CO2/year
Home Energy         :   2397.12 kg CO2/year
-----
Total Footprint      :   3609.14 kg CO2/year
-----

Indian Average : 1900 kg CO2/year
Global Average : 4000 kg CO2/year

Status : Moderate footprint.

=====

SUGGESTIONS
=====

Reduce meat consumption.
Potential savings : 291.50 kg CO2/year

Use energy-efficient appliances.
Possible reduction : 479.42 kg CO2/year

Save report to file? (1 = Yes, 0 = No) : 1

Report saved successfully.

*****
Thank you for using the calculator!
*****

=== Code Execution Successful ===
```

Emission Factors & Conclusion

Emission Factor Reference Table

Source / Category	#define Constant	Emission Factor
Bus	BUS_PER_KM	0.089 kg CO2 per km
Bike / Motorbike	BIKE_PER_KM	0.103 kg CO2 per km
Train	TRAIN_PER_KM	0.041 kg CO2 per km
Non-Veg Diet	DIET_NONVEG	3.3 kg CO2 per day
Mixed Diet	DIET_AVERAGE	2.5 kg CO2 per day
Vegetarian Diet	DIET_VEGETARIAN	1.7 kg CO2 per day
Electricity	ELECTRICITY_PER_KWH	0.82 kg CO2 per kWh (India grid)
LPG Gas	LPG_PER_KG	2.983 kg CO2 per kg
Indian Average	avgIndian	1,900 kg CO2/year
Global Average	avgGlobal	4,000 kg CO2/year

Status Classification

Annual CO2 Footprint	Status Verdict
Total < 1,900 kg/year	Excellent — Below Indian Average
1,900 – 4,000 kg/year	Moderate — Between Indian & Global Average
Total > 4,000 kg/year	High — Exceeds Global Average

Conclusion

The **Carbon Footprint Calculator** project successfully achieves its objective of providing a practical, user-friendly tool for estimating personal CO2 emissions. By implementing three well-structured OOP classes — **TransportData**, **DietData**, and **HomeEnergyData** — the project demonstrates a clear and professional application of C++ programming fundamentals.

The program applies real Indian emission factors, benchmarks results against national and global averages, and generates targeted, quantified reduction suggestions. The built-in file export feature using **ofstream** adds practical value, allowing users to save and revisit their carbon reports.

This project serves as a meaningful contribution to **UN SDG Goal 13: Climate Action**, demonstrating how even beginner-level C++ can be used to build software with real-world environmental impact and educational value.